

# AREENA MAHEK

## Forward Deployed Engineer

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### PROFESSIONAL SUMMARY

Forward Deployed Engineer with 3 years of experience building and deploying AI-powered, enterprise-scale solutions in production. Works closely with clients to turn complex business problems into scalable systems – integrating SAP S/4HANA and SAP Business One, and implementing LLM-driven workflows including RAG-based chatbot solutions for predictive maintenance. Known for turning ambiguous requirements into practical, high-impact solutions.

### RELEVANT EXPERIENCE

#### Oxmaint Inc. | Lead Forward Deployed Engineer | Sunnyvale, CA - Remote

Aug 2025 – Present

- Delivered client-facing deployments across 5+ enterprise clients, supporting 20+ active users per client and integrating SAP S/4HANA and AI workflows to enable real-time operational insights.
- Owned end-to-end delivery of enterprise solutions, translating business requirements into scalable architectures and accelerating client onboarding.
- Integrated a RAG-based predictive analytics pipeline into a production chatbot, leveraging historical asset data and processing 100K+ maintenance records to surface downtime insights and enable proactive maintenance decisions.
- Engineered enterprise integrations across SAP S/4HANA (BTP, OData), SAP Business One, and QuickBooks, streamlining financial and operational workflows.
- Designed and deployed AI-driven automation workflows (n8n, GPT, Claude), reducing onboarding time by 40% and improving operational efficiency.
- Built full-stack enterprise solutions (React, Next.js, TypeScript, Golang), including APIs and scalable backend services tailored to client requirements.
- Translated ambiguous client requirements into production-ready solutions by defining system architecture, data flows, and integration strategies.

#### DePaul University | Research Assistant | Chicago, IL

##### Scalable Data Management Service as a Microservice – ByteBridge

Jan 2025 – June 2025

- Architected a modular microservice-based data platform using Django and REST APIs, improving research data accessibility and visualization by 70%.
- Developed frontend interfaces (HTML, CSS, JavaScript) to enable efficient data interaction and visualization for research workflows.
- Ensured system reliability through version control, code reviews, and unit testing, improving maintainability and deployment stability.

#### Interactive Out-of-Distribution (OoD) Data Visualization Dashboard

Aug 2024 – Dec 2024

- Built an interactive visualization dashboard using Python, Trame, and Vuetify (Vue.js) to support anomaly detection in high-dimensional medical imaging data (LIDC dataset).
- Implemented threshold-based filtering and scoring mechanisms to classify OoD samples, improving interpretability and model debugging workflows.

#### DePaul University | Graduate Assistant-Tutoring Mentor | Chicago, IL

Sept 2023 – June 2025

- Mentored 200+ students across Python, Java, SQL, distributed systems, and full-stack development, improving academic performance by 95%.
- Guided students in debugging, optimizing, and designing scalable applications, strengthening practical understanding of system design and algorithms.

### SKILLS

- AI & LLM Systems:** OpenAI, Anthropic Claude, RAG, Vector DBs (Milvus, MongoDB), NLP (Hugging Face, NLTK)
- Backend & APIs:** Golang, Python (Django, Flask), REST APIs, Microservices, System Design, Trame
- Frontend:** React.js, Next.js, TypeScript, JavaScript (ES6+), Vue.js, Tailwind CSS
- Enterprise Integrations:** SAP S/4HANA (BTP, OData), SAP Business One, QuickBooks
- Cloud & Infrastructure:** AWS (S3, EMR, EC2, Athena), Azure, Docker, Linux
- Data & Analytics:** SQL (PostgreSQL, MySQL, Oracle), Pandas, NumPy, Scikit-learn, Tableau, Power BI
- Engineering & DevOps:** Git, JWT, OAuth2, Unit Testing, Data Structures & Algorithms

### EDUCATION

#### DePaul University | Chicago, IL, USA

Jan 2023 – June 2025

M.S. in Computer Science with Data Science | GPA: 3.96/4.0

#### Stanley College of Engineering and Technology for Women | India

MTech. in Computer Science | GPA: 3.6/4.0

Dec 2020 – Dec 2022

#### Muffakhham Jah College of Engineering and Technology | India

Aug 2016 – Sept 2020

B.E. in Information Technology | GPA: 3.75/4.0

### HONORS & AWARDS

- UPE Academic Excellence Honor; M.S. (CS, Data Science) with Distinction (2024)
- MTech College Topper (2020–2022); B.E University Topper (2016 – 2020)

### LEADERSHIP & ENGAGEMENT

- Workshop Instructor – Trame (Python Visualization)** | DePaul University | Feb 2025 – Delivered hands-on session on building interactive data visualization apps using Trame & Vuetify (Vue.js).
- Speaker – Research Symposium** | DePaul University | Mar 2025 – Presented OoD visualization work for medical imaging and model interpretability.
- Graduate Student Ambassador** | DePaul University | Aug 2023 – Mar 2025 – Guided student onboarding and represented university at international outreach events.